

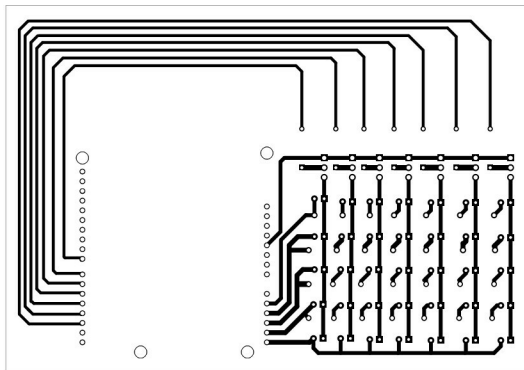


Fig. 2: Actual-size PCB layout of the 7x5 LED matrix using Arduino

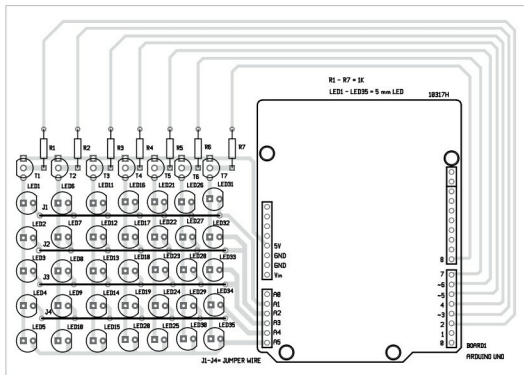


Fig. 3: Component layout of the PCB

to another set of pins on Arduino board, represented as column pins in the code.

Multiplexing LED matrix.

Multiplexing is employed to operate LED matrices. By using this technique, different LEDs can have common row and column lines. When one

row is activated, one lead each of all LEDs in that row gets the required voltage. In order to make a single LED in this row to glow, the corresponding column line is activated.

Making all LEDs glow in a particular row by activating all column transistors may cause overload for Arduino. To avoid that, the column transistors are switched between on and off states so fast that, in effect, all LEDs seem to glow continuously—a phenomenon called persistence of vision.

Similarly, to display a character, the LEDs required to display the character are activated and deactivated one by one in high speed, tricking the eye to interpret it as a continuous glow of LEDs, thus seeing the required character glow continuously.

Collectors of all transistors are connected to 5V supply from Arduino Uno board, emitter pins are connected to each column of the LED matrix and bases of these transistors are connected to digital pins 2, 3, 4, 5, 6, 7 and 8 of Uno board through current-limiting resistors

(R1-R8). Analogue input pins are used as digital pins in this project to provide ground potential for the LED terminals.